ModelIt!

Teacher's Guide

Can a River Eat a Mountain?

The Slow but Powerful Force of Water on Rock

Quick Reference

Grade Level	Grade 4
NGSS Standard	4-ESS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: 4-ESS2-1

Make observations and/or measurements to provide evidence of the effects of weathering or the rate of erosion by water, wind, ice, and vegetation.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students make observations and gather evidence about how water flow speed and rock hardness affect erosion rate.
Disciplinary Core Idea	**ESS2.A Earth Materials and Systems** — Rainfall helps to shape the land and affects the types of living things found in a region. Water, ice, wind, living organisms, and gravity break rocks, soils, and sediments into smaller particles and move them around.
Crosscutting Concept	**Cause and Effect** — Students identify cause and effect relationships between water speed, rock hardness, and the rate of erosion.

CAST Testing Alignment

MULTIPLE CHOICE:

Two rivers flow at the same speed. River A flows over soft sandstone and River B flows over hard granite. Which river will carve a deeper channel in the same amount of time?

CONSTRUCTED RESPONSE:

Explain how the Grand Canyon was formed using the words erosion, water flow, and rock layers in your answer.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Erosion is one of the most powerful forces shaping Earth's surface. While it works slowly compared to human lifetimes, over geological time it has carved the Grand Canyon, worn down entire mountain ranges, and created fertile river deltas.

How Water Erodes Rock

Moving water erodes rock through three main processes: hydraulic action (the force of water itself loosening rock), abrasion (sand and pebbles carried by water grinding against rock like sandpaper), and dissolution (water dissolving minerals in certain types of rock). Faster water has more kinetic energy and can erode far more effectively than slow water.

The Grand Canyon Story

The Grand Canyon is approximately 277 miles long, up to 18 miles wide, and over a mile deep. It was carved primarily by the Colorado River over 5-6 million years. The canyon's colorful rock layers span nearly 2 billion years of geological history. The oldest rocks at the bottom (Vishnu Basement Rocks) are about 1.84 billion years old.

Rock Hardness and Erosion

The Mohs hardness scale rates minerals from 1 (talc) to 10 (diamond). Soft rocks like limestone and sandstone erode much faster than hard rocks like granite and quartzite. This differential erosion is why canyon walls often have a staircase pattern — hard layers stick out while soft layers are carved deeper. It also explains why some mountains are taller than others.

Erosion and Human Life

Erosion creates fertile farmland by depositing nutrient-rich sediment in river valleys and deltas. The Nile Delta, Mississippi Delta, and Ganges Delta are among the most productive agricultural areas on Earth — all created by erosion. However, erosion also threatens structures, roads, and homes built near rivers and coastlines. Managing erosion is a major engineering challenge.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify water flow speed, rock hardness, and erosion rate as the key components of the weathering and erosion system.

E **Establish Relationships**
Students determine that water flow speed positively affects erosion rate (faster water = more erosion), while rock hardness negatively affects erosion rate (harder rock = less erosion).

V **Visualize & Model**
Students build a model showing how fast-flowing water interacts with different types of rock to produce different erosion rates.

E **Evaluate Outcomes**
Students run gentle stream, raging river, and soft rock scenarios to observe how changing water speed and rock type affects erosion.

R **Revise & Extend**
Students add wind speed, ice formation, or plant roots to explore other forces that break down rock.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"This canyon is a MILE deep. And the only thing that carved it was water. How is that possible?"*

Watch For

Show a dramatic photo of the Grand Canyon. Let students express wonder.

Slide 2: Learning Objectives

2 min

Say: *"Today we are going to figure out how something soft can destroy something hard."*

Watch For

Have students read objectives aloud. Pre-teach weathering and erosion.

Slide 3: Big Question

3 min

Say: *"You drink water. You swim in water. You wash your hands with water. Can it really eat a mountain?"*

Watch For

Quick poll: How many students think water can break rock? Discuss initial ideas.

Slide 4: LEVER Framework

2 min

Say: *"We will build a model to test whether water can really destroy rock."*

Watch For

Preview LEVER steps. Connect to student curiosity.

Slide 5: Activity 1: Components

7 min

Say: *"What affects how fast rock wears away? Think about the water AND the rock."*

Watch For

Guide component sorting. Discuss why water speed and rock hardness come from outside the system.

Slide 6: Activity 2: Connections

7 min

Say: *"Faster water means MORE erosion — that is positive. But harder rock means LESS erosion — that is negative!"*

Watch For

Help students draw arrows. This lesson has one positive and one negative relationship.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us see what happens when a raging river meets soft rock!"*

Watch For

Students predict, then run scenarios. Compare gentle stream vs raging river.

Slide 8: Discoveries

4 min

Say: *"Now you know the answer — yes, a river CAN eat a mountain. It just takes millions of years!"*

Watch For

Connect model results to Grand Canyon. Discuss how time changes everything.

Slide 9: STEM Challenge

8 min

Say: *"A town needs your help — the river is eating their riverbank! Design a solution."*

Watch For

Distribute materials. Groups design erosion-proof riverbank solutions.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Water Flow Speed, Rock Hardness
- **INTERNAL:** Erosion Rate

Relationships

- **Water Flow Speed to Erosion Rate:** POSITIVE (+) — Faster water has more energy to break apart rock and carry away pieces. A raging river erodes much faster than a gentle stream.
- **Rock Hardness to Erosion Rate:** NEGATIVE (-) — Harder rock resists being broken apart. Soft sandstone erodes easily while hard granite takes much longer to wear away.

Key Discoveries

- Faster water has more energy and can break apart and carry away larger pieces of rock
- Softer rock erodes much faster than harder rock, which is why canyons have uneven walls
- Even very slow water can wear away the hardest rock if given enough time — millions of years
- The Grand Canyon was carved by the Colorado River over about 5-6 million years

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Water is too soft to break rock"	While individual water molecules are tiny, moving water has tremendous energy. A fast-flowing river carries sand and pebbles that act like sandpaper, grinding rock away. Water also seeps into cracks and freezes, expanding and breaking rock apart. Over millions of years, water has carved the deepest canyons and worn down the tallest mountains on Earth.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Erosion only happens during floods or storms"	Erosion happens every single day, even when you cannot see it. Every raindrop that hits soil, every stream that flows over rock, and every wave that hits a beach is causing erosion. Major storms cause dramatic erosion, but the slow, constant erosion happening every day actually shapes more of the landscape over geological time.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The Grand Canyon was formed quickly by a huge flood"	The Grand Canyon was carved primarily by the steady flow of the Colorado River over 5-6 million years, not by a single catastrophic event. While floods did speed up erosion at times, the canyon is the result of constant, slow erosion over an almost unimaginable span of time. The deepest rock layers exposed at the bottom are nearly 2 billion years old.